An explainable algorithm for credit risk prediction based on contrast patterns

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Abstract. Resumen

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1 Introduction

Credit risk prediction involves determining whether an applicant will default on a future financial obligation [1], and is therefore a fundamental task in the financial sector. Credit risk prediction is commonly as a supervised classification problem; the goal is to assign each applicant to a class (e.g., default or non-default), rather than estimating a continuous predictive risk value. Credit risk prediction directly impacts the stability of banks and financial institutions, where accurate and explainable predictions are necessary to ensure regulatory compliance [2, 3]. Increasingly, financial institutions have implemented evaluations and management processes that employ pattern recognition and machine learning algorithms for credit risk prediction [4, 5]. As these predictors become more widely adopted, improving default prediction enables more accurate credit risk prediction and reduces losses [6–8].

Recent research in credit risk prediction has prioritized improving the performance of credit risk prediction algorithms based on machine learning and pattern recognition through the use of oversampling or undersampling methods, thereby reducing algorithmic bias towards the majority class [9–13]; the use of feature selection methods for identifying the most relevant features of a dataset to improve the performance of prediction algorithms [1, 14–16]; and the use of explainable AI techniques [10, 17–19] that allow users to comprehend and trust the predictions provided by credit risk prediction algorithms, which makes the predictions of the algorithm to comply with regulations.

Machine learning algorithms have demonstrated good predictive performance, and decision trees are among the most effective for credit risk prediction [4, 20–25]. However, the need for explanations remains a significant limitation, particularly in the financial sector, as financial institutions and regulators require

justification for their decisions [26, 27]. Several authors have addressed the lack of explainability in credit risk prediction [17, 28, 29].

Explainability techniques, also known as XAI (eXplainable Artificial Intelligence), aim to make machine learning prediction processes transparent and understandable to humans. These techniques are important in areas such as credit risk prediction, where justifying and explaining the predictions is essential in enabling financial institutions to make more informed decisions. Several recent studies have applied SHAP [30] and LIME [31] to credit risk prediction to generate explanations for individual predictions [32–34]. SHAP assigns to each feature a value that represents how much it contributed to the prediction. These values help explain the role each feature played in the prediction. LIME builds a simple interpretable predictor that approximates the behaviour of the original predictor by modifying a specific instance and observing the changes in the prediction. The result is an explanation that highlights the features contributing most to the prediction, showing how each affects the prediction in a positive or negative direction. However, these techniques focus on individual features and often overlook how several features jointly contribute to the prediction.

In credit risk prediction, a predictor's output depends on multiple features rather than individual features. Therefore, explainable predictors should consider multiple features to explain the prediction [18, 35]. In response to this need, this article proposes a credit risk prediction algorithm that generates explanations based on contrast patterns, which are conjunctions of conditions of the form $feature\#value$ that are frequent in a class but rarely appear in the opposite class. Since these patterns can be naturally extracted using decision-tree-based miners, we propose using decision-tree-based predictors to predict credit risk, which have reported competitive credit risk predictions compared to the most successful approaches. In this way, the proposed algorithm leverages the affinity and synergy between these two decision-tree-based algorithms.

Unlike other explainable algorithms that merely highlight the importance of individual features, our method explains predictions based on contrast patterns present in the training data and shared with the new applicant. Experiments conducted on both public and private financial datasets show that the proposed algorithm achieves competitive predictive accuracy while offering instance-level explanations based on the contrast patterns shared with other clients in the class, unlike the current state-of-the-art, which explains based on the single features contributing most to the prediction.

The remainder of this paper is organised as follows. Section 2 presents the related work. Section 3 describes the proposed algorithm. Section 4 presents our experimental results and a discussion about them. Finally, Section 5 presents our conclusions and future work directions.

2 Related works

Credit risk prediction has been extensively studied in the literature using a wide range of pattern recognition and machine learning algorithms [3, 7, 8, 10, 14, 15, 28, 29, 33, 36–48]. In this section, we review the most recent works on credit risk prediction, with a focus on approaches that either prioritize achieving the highest accurate credit risk prediction or incorporate explainability for credit risk prediction. The works considered in this review are those that utilize publicly available datasets, commonly used in the literature and accessible online, which allows a comparison against our proposal. Accordingly, this section is divided into two parts: works that focus solely on accuracy without offering explanations, and works that integrate explainability into their predictors.

2.1 Predictors for Credit Risk Prediction

Recent works on credit risk prediction have focused on developing predictive algorithms to improve performance metrics, such as accuracy, area under the curve (AUC), Type II error, or F1-score.

In [38], the authors proposed a stacked ensemble for credit risk prediction that combines Random Forests (RF), Logistic Regression (LR), and Convolutional Neural Networks (CNN). The results of these predictors are used as input features for a Multilayer Perceptron (MLP), which learns to combine them to produce the prediction. The authors used SMOTEENN [49] to address class imbalance, and the stacked ensemble is compared against CNN, LR, MLP, and RF. After applying SMOTE-ENN to address class imbalance, the stacked ensemble consistently outperformed the individual predictors, achieving accuracies near 96%.

A comparative study of different predictors for credit risk prediction is presented in [46], including LR, RF, and Artificial Neural Networks (ANN). In this study, SMOTE [50] was employed to balance the training dataset, leading to significant improvements in the performance metrics of all predictors. Among them, RF achieved the best results, with an accuracy of 98%.

In [47], Deep Convolutional Neural Networks (DCNN) and Support Vector Regression (SVR) were compared for credit risk prediction. The application of SMOTE contributed to improved predictive performance in both predictors. Using SMOTE to balance the dataset, SVR achieved an accuracy of 98% and F1-score of 87%.

In [51], the authors present the Level-Wise Feature-Guided Cascading Ensembles (LFGCE) model, which integrates hierarchical feature selection with a cascading ensemble learning scheme. To evaluate its performance, several predictors were employed, including RF, Support Vector Machines (SVM), LR, Decision Trees (DT), ANN, Linear Discriminant Analysis (LDA), and XGBoost.

Among these, XGBoost achieved the best results in both feature selection and prediction, with an AUC of 94%. However, it is important to note that the study does not address class imbalance, as no data balancing methods were applied.

Most of the reviewed studies emphasise improving predictive performance through class balancing methods to mitigate data imbalance. Several predictors yield good results in credit risk prediction. However, the reported performance metrics (such as accuracy, AUC, and F1-score) vary significantly on the chosen predictors and balancing methods. Overall, the authors consistently point out that addressing class imbalance is crucial for achieving reliable predictions, with many studies reporting high accuracy values, often exceeding 90%.

2.2 Explainable Predictors for Credit Risk Prediction

Incorporating explainability into credit risk prediction has been a growing area of study. These studies aim to achieve robust predictive performance while prioritising the explainability of predictions.

A study by [43] addresses credit risk prediction by comparing the predictive performance of XGBoost, RF, Least Squares SVM (LSSVM), and a CNN. After applying SMOTE to balance the dataset, XGBoost achieved the best predictive accuracy, reaching 92%. To enhance predictor explainability, SHAP analysis was applied to quantify the individual contributions of each feature. From the predictors evaluated, XGBoost performed best in terms of predictive accuracy and also provided explanations through SHAP values.

In [33], the authors present an ensemble framework that combines RF and XGBoost using probability-weighted averaging. The framework was evaluated using eight predictors, including LR, DT, Gradient Boosting (GB), LightGBM (LGBM), MLP, Naive Bayes (NB), and XGBoost. Their proposed ensemble significantly outperformed all others, achieving an AUC of 97%. SHAP was employed to explain the predictions, which allowed them to determine the contribution of each feature.

In [29], XGBoost is used as a predictor, combined with SHAP, LIME, and K-means to enhance explainability, achieving an AUC of 78%. SHAP and LIME are used to generate feature-contribution vectors for each training instance, which are grouped using K-means. For explaining the prediction of a new instance, its feature-contribution vector is calculated and assigned to the nearest cluster, and the prediction explanation is provided as the cluster's average vector.

In [3], the authors present a comparative analysis of the performance of boosting algorithms, including LightGBM, XGBoost, CatBoost, and AdaBoost. Additionally, SMOTE-Tomek [52] is employed to balance the dataset. The results indicate that LightGBM achieves the best performance with 95% accuracy and 98% AUC, outperforming the other boosting methods. SHAP is applied to

quantify the contribution of each feature to the prediction.

In [28], a multilayer dynamic graph neural network (DYMGNN), proposed for credit risk prediction. The authors compare their approach against several baselines: LR, XGBoost, and DNN. Unlike many related studies, no resampling or class balancing methods are applied. Experimental results show that the DYMGNN achieves the best results with 81% accuracy and 83% F1-score. For explainability, SHAP is used to identify the features that contribute most to the predictions.

In [48], the authors propose an explainable two-stage technique to explain the results of credit risk prediction. In the first stage, SHAP is applied to analyse the importance of features according to their positive or negative impact on the prediction. In the second stage, they use a genetic algorithm to generate counterfactual explanations by perturbing the features according to this categorisation, thereby identifying the minimum necessary changes to modify the prediction. This technique was evaluated using multiple predictors, including SVM, XGBoost, MLP, and LSTM, with an accuracy of 90% for the employed predictors. The results demonstrate that the features modified most frequently closely align with the features identified as most relevant by SHAP.

In [53], the authors compare the performance of several Deep Learning (DL) architectures for credit risk prediction, including Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRU), and Graph Neural Networks (GNN). To address class imbalance, they apply SMOTEENN. The experimental results show that GRU outperformed all other predictors, achieving an accuracy of 91%. To explain the prediction, SHAP is used to quantify the importance of features and their contributions to the predictions.

In [7], the authors propose an approach for credit risk prediction on imbalanced datasets by integrating the TabTransformer model with a weighted loss function. This approach helps mitigate the bias toward the majority class, improving predictive performance on minority classes, achieving 91% accuracy and an AUC of 93%. SHAP is employed to evaluate feature importance and identify those features that contribute most to the predictions.

In [54], the authors propose a robust framework for credit risk prediction on imbalanced datasets by introducing the K-SMOTEENN method, which integrates oversampling and noise-filtering methods with a stacking ensemble. The ensemble combines Random Forest, XGBoost, and LightGBM, resulting in a predictive performance of 92% F1-score, and 96% AUC. SHAP is used to analyse and explain the contributions of individual features to the prediction.

As discussed in the above works, credit risk prediction has been addressed through a wide range of predictors, balancing methods, feature selection methods, and evaluated through multiple performance metrics. Nevertheless, accurately detecting defaults remains a significant challenge, as no single method has consistently outperformed others across different scenarios. Moreover, the most commonly used explainability approaches, such as SHAP and LIME, while helpful in assessing the individual contribution of features, fail to capture feature value combinations that characterise different types of applicants. This limitation is particularly relevant, since financial institutions increasingly demand not only high predictive accuracy but also clear and case-specific justifications, both to meet regulatory requirements and to improve communication with applicants.

3 Proposed explainable algorithm

As discussed in the previous section, recent research has increasingly focused on explaining credit risk predictions. Techniques such as SHAP and LIME, commonly used for explaining credit risk predictions, assign importance scores to individual features; however, they do not account for combinations of feature values when explaining a prediction. This limitation highlights the need for approaches that not only assess the contribution of individual features when explaining predictions, but also combinations of them.

The idea behind our algorithm is to integrate tree-based predictors with contrast pattern mining algorithms. Unlike traditional explainers for credit risk prediction, which are typically formulated as post-hoc explanation techniques based on feature attribution methods (e.g., SHAP or LIME), our algorithm maintains a separation between the prediction and explanation processes, adopting contrast pattern mining as an explanatory mechanism. Furthermore, both the predictor and the explanation are grounded in tree-based structures, allowing the explanations to reflect the same decision structure used in the prediction.

As mentioned previously, credit risk prediction depends on combinations of feature values rather than individual features considered in individually. For instance, a 'high debt-to-income ratio' may only indicate default when combined with 'low job stability'. While existing explanation techniques for credit risk prediction typically evaluate features individually, our algorithm represents combinations of feature values through contrast patterns. These patterns provide explanations based on conjunctions of feature conditions, similar to those used by tree-based predictors.

Our choice of tree-based predictors due not only to their success in state-of-the-art credit risk prediction [38], but also to their alignment with contrast patterns. This compatibility enables a transition between prediction and explanation. In particular, a decision tree can be interpreted as a hierarchy of decision rules, where each path from the root to a leaf node corresponds to a conjunction

of conditions the form $feature\#value$. Similarly, contrast patterns are expressed as conjunctions of conditions $feature\#value$.

By employing a contrast pattern mining algorithm compatible with this representation, a direct and consistent correspondence is maintained between the predictor and the generated explanations.

3.1 First stage

Because the datasets for credit risk prediction are imbalanced, and following the results reported in [55], we propose to apply SMOTENC [50, 56] to the training set to address class imbalance before training a predictor or extracting contrast patterns.

For prediction, we propose using the FT4cip predictor [57], a tree-based predictor capable of handling mixed and imbalanced datasets, which has reported good results with this type of dataset making it suitable for credit risk prediction. To construct the explanation, we propose using contrast patterns extracted from similar instances within each class. However, instances within the same class may exhibit different combinations of feature values, forming distinct subgroups within the class. Therefore, we propose extracting contrast patterns from clusters generated within each class of the dataset, allowing the identification of patterns for each subgroup.

To generate clusters, we first split the dataset into classes (defaulter and non-defaulter). Within each class, we propose using the Kmodes algorithm [58, 59]. This choice is motivated by the need to group instances with similar feature-value combinations, particularly in mixed-type data. To determine the number of clusters in each class, we use the Silhouette coefficient [60, 61], ensuring that the resulting clusters capture well-defined groupings within each class. This clustering step enables the extraction of contrast patterns that reflect localized structures within each class.

For mining contrast patterns, we use CPMining, introduced in the PBC4cip algorithm [62]. This contrast pattern miner is supervised, meaning that it requires labeled data and explicitly exploits differences between classes to identify discriminative patterns. Thus, we propose applying it on each cluster of a given class against the other class, rather than using the entire class as a whole. This design choice is motivated by two main factors. First, by comparing each cluster against the opposite class, the extracted patterns remain class-discriminative, which is consistent with the supervised nature of the mining algorithm. Second, operating at the cluster level allows capturing more specific combinations of feature values that characterize subgroups within the class, instead of producing overly general patterns. As a result, the generated contrast patterns are both cluster-specific and class-discriminative, capturing more specific contrast

patterns while maintaining explainability.

After mining, we propose a filtering process to remove redundant contrast patterns. Following the filtering strategy used in CPMining, redundancy is identified by comparing pairs of patterns based on their conditions. Given two patterns P_1 and P_2, P_1 is considered more general than P_2 if its set of conditions is a subset of those defining P_2, in which case P_2 is more specific. Applied at the cluster level, this process effectively removes redundant contrast patterns while preserving the most representative and informative ones, thereby enhancing the explainability of the extracted patterns and reducing their overall number.

For example, consider the next contrast patterns extracted from the German Credit Data dataset [63]:

P_1 : Credit Duration $\leq$ 15 months **and** Property=real estate **and** Foreign Worker=yes

and the more specific contrast pattern

P_2 : Credit Duration $\leq$ 15 months **and** Property=real estate **and** Foreign Worker=yes **and** Checking Account $\neq$ low

In this case P_1 is more general than P_2. If both contrast patterns cover the same instances within the cluster, the second contrast pattern does not provide additional information, since it introduces an extra condition while describing the same subset of instances. Therefore, only the more general contrast pattern is retained.

Additionally, contrast patterns whose support in the opposite class is greater than zero are removed, ensuring that the final set contains only contrast patterns that characterize the cluster while distinguishing it from the other class.

References

[1] Jianxin Zhu, Xiong Wu, Lean Yu, and Xiaoming Zhang. "A hybrid clustering and boosting tree feature selection (CBTFS) method for credit risk assessment with high-dimensionality". In: *Technological and Economic Development of Economy* (2025), pp. 1–33. DOI: 10.3846/tede.2025.23060.

[2] Xiaosong Zhao, Yong Liu, and Qiangfu Zhao. "Improved LightGBM for extremely imbalanced data and application to credit card fraud detection". In: *IEEE Access* (2024). DOI: 10.1109/ACCESS.2024.3487212.

[3] Nhat Nguyen and Duy Ngo. "Comparative analysis of boosting algorithms for predicting personal default". In: *Cogent Economics & Finance* 13.1 (2025), p. 2465971. DOI: 10.1080/23322039.2025.2465971.

[4] Patrick Weber, K Valerie Carl, and Oliver Hinz. "Applications of explainable artificial intelligence in finance—a systematic review of finance, information systems, and computer science literature". In: *Management Review Quarterly* 74.2 (2024), pp. 867–907. DOI: 10.1007/s11301-023-00320-0.

[5] Wei Jie Yeo, Wihan Van Der Heever, Rui Mao, Erik Cambria, Ranjan Satapathy, and Gianmarco Mengaldo. "A comprehensive review on financial explainable AI". In: *Artificial Intelligence Review* 58.6 (2025), pp. 1–49. DOI: 10.1007/s10462-024-11077-7.

[6] Sichong Lu, Yi Su, Xiaoming Zhang, Jiahui Chai, and Lean Yu. "LLM-infused bi-level semantic enhancement for corporate credit risk prediction". In: *Information Processing & Management* 62.4 (2025), p. 104091. DOI: 10.1016/j.ipm.2025.104091.

[7] Kristoko Dwi Hartomo, Christian Arthur, and Yessica Nataliani. "A Novel Weighted Loss TabTransformer Integrating Explainable AI for Imbalanced Credit Risk Datasets". In: *IEEE Access* (2025). DOI: 10.1109/ACCESS.2025.3541878.

[8] Tao Yu, Wei Huang, Xin Tang, and Duosi Zheng. "A hybrid unsupervised machine learning model with spectral clustering and semi-supervised support vector machine for credit risk assessment". In: *PloS one* 20.1 (2025), e0316557. DOI: 10.1371/journal.pone.0316557.

[9] I Nyoman Mahayasa Adiputra, Pei-Chun Lin, and Paweena Wanchai. "The Effectiveness of Generative Adversarial Network-Based Oversampling Methods for Imbalanced Multi-Class Credit Score Classification". In: *Electronics* 14.4 (2025), p. 697. DOI: 10.3390/electronics14040697.

[10] Shaukat Ali Shahee and Rujavi Patel. "An Explainable ADASYN-Based Focal Loss Approach for Credit Assessment". In: *Journal of Forecasting* (2025). DOI: 10.1002/for.3252.

[11] Kianeh Kandi and Antonio García-Dopico. "Enhancing Performance of Credit Card Model by Utilizing LSTM Networks and XGBoost Algorithms". In: *Machine Learning and Knowledge Extraction* 7.1 (2025), p. 20. DOI: 10.3390/make7010020.

[12] Saad M Darwish, Amr Ibrahim Salama, and Adel A Elzoghabi. "Intelligent approach to detecting online fraudulent trading with solution for imbalanced data in fintech forensics". In: *Scientific Reports* 15.1 (2025), pp. 1–19. DOI: 10.1038/s41598-025-01223-8.

[13] Ya-Han Hu, Chih-Fong Tsai, and Pei-Ting Wang. "Combining multiple data resampling methods and classifier ensembles for better financial distress prediction: homogeneous and heterogeneous approaches". In: *Annals of Operations Research* (2025), pp. 1–22. DOI: 10.1007/s10479-025-06706-5.

[14] Ines Gasmi, Sana Neji, Salima Smiti, and Makram Soui. "Features Selection for Credit Risk Prediction Problem". In: *Information Systems Frontiers* (2025), pp. 1–20. DOI: 10.1007/s10796-024-10559-x.

[15] Shujie Zou, Zhiming Cai, Chiawei Chu, Zefeng Zhao, Haohao Cai, Ning Shen, and Jie Ren. "Variable Selection and Fusion Sampling of Unbalanced

Data in Credit Risk Assessment". In: *Computational Economics* (2025), pp. 1–31. DOI: 10.1007/s10614-025-10943-y.

[16] Huanjing Wang, Qianxin Liang, John T Hancock, and Taghi M Khoshgoftaar. "Feature selection strategies: a comparative analysis of SHAP-value and importance-based methods". In: *Journal of Big Data* 11.1 (2024), p. 44. DOI: 10.1186/s40537âĂŠ024âĂŠ00905âĂŠw.

[17] Shuoyan Lin, Dandan Song, Boyi Cao, Xin Gu, and Jiazhan Li. "Credit risk assessment of automobile loans using machine learning-based SHapley Additive exPlanations approach". In: *Engineering Applications of Artificial Intelligence* 147 (2025), p. 110236. DOI: 10.1016/j.engappai.2025.110236.

[18] Yan Zhang, Lin Chen, and Yixiang Tian. "A Method for Evaluating the Interpretability of Machine Learning Models in Predicting Bond Default Risk Based on LIME and SHAP". In: *arXiv preprint arXiv:2502.19615* (2025). DOI: 10.48550/arXiv.2502.19615.

[19] Seongil Han and Haemin Jung. "NATE: Non-pArameTric approach for Explainable credit scoring on imbalanced class". In: *PloS one* 19.12 (2024), e0316454. DOI: 10.1371/journal.pone.0316454.

[20] Victor Chang, Qianwen Ariel Xu, Shola Habib Akinloye, Vladlena Benson, and Karl Hall. "Prediction of bank credit worthiness through credit risk analysis: an explainable machine learning study". In: *Annals of Operations Research* (2024), pp. 1–25. DOI: 10.1007/s10479-024-06134-x.

[21] Idowu Aruleba and Yanxia Sun. "Effective credit risk prediction using ensemble classifiers with model explanation". In: *IEEE Access* (2024). DOI: 10.1109/ACCESS.2024.3445308.

[22] Xinjian Zhang, Yuting Zhang, Xuefeng Liu, and Ruping Wang. "Blockchain-Based Intelligent Risk Management Decision Support System for Supply Chain Financing". In: *International Journal of Intelligent Information Technologies (IJIIT)* 21.1 (2025), pp. 1–24. DOI: 10.4018/IJIIT.369153.

[23] Meng Xia, Zhijie Wang, Xingyu Lan, Wanan Liu, and Jiawei Wu. "Confidence-driven under-sampling decision forest for imbalanced credit scoring". In: *Engineering Applications of Artificial Intelligence* 147 (2025), p. 110278. DOI: 10.1016/j.engappai.2025.110278.

[24] Chenxi Wu. "Credit risk unveiled: Decision trees triumph in comparative machine learning study". In: *Applied and Computational Engineering* 79 (2024), pp. 174–181. DOI: 10.54254/2755-2721/79/20241613.

[25] Yao Zou, Meng Xia, and Xingyu Lan. "Interpretable credit scoring based on an additive extreme gradient boosting". In: *Chaos, Solitons & Fractals* 194 (2025), p. 116216. DOI: 10.1016/j.chaos.2025.116216.

[26] MN Sowmiya, S Jaya Sri, S Deepshika, and G Hanushya Devi. "Credit Risk Analysis using Explainable Artificial Intelligence". In: *Journal of Soft Computing Paradigm* 6.3 (2024), pp. 272–283. DOI: 10.36548/jscp.2024.3.004.

[27] Vishnu Ravi, Vineet Kumar Srivastava, Maninder Pal Singh, Ravi Kumar Burila, Nikhil Kassetty, Padma Naresh Vardhineedi, Venkata Reddy

Pasam, Nuzhat Noor Islam Prova, and Indrajit De. "Explainable AI (XAI) for Credit Scoring and Loan Approvals". In: *arXiv* (2025).

[28] Sahab Zandi, Kamesh Korangi, María Óskarsdóttir, Christophe Mues, and Cristián Bravo. "Attention-based dynamic multilayer graph neural networks for loan default prediction". In: *European Journal of Operational Research* 321.2 (2025), pp. 586–599. DOI: 10.1016/j.ejor.2024.09.025.

[29] Xinyu Sun, Jiayu Liu, and Yan Zhang. "Enhancing Credit Risk Prediction through an Ensemble of Explainable Model". In: *Journal of Systems Science and Systems Engineering* (2025), pp. 1–22. DOI: 10.1007/s11518-025-5663-y.

[30] Scott M Lundberg and Su-In Lee. "A unified approach to interpreting model predictions". In: *Advances in neural information processing systems* 30 (2017).

[31] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "" Why should i trust you?" Explaining the predictions of any classifier". In: *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*. 2016, pp. 1135–1144. DOI: 10.1145/2939672.2939778.

[32] Rubén García-Céspedes, Francisco J Alias-Carrascosa, and Manuel Moreno. "On Machine Learning models explainability in the banking sector: the case of SHAP". In: *Journal of the Operational Research Society* (2025), pp. 1–13. DOI: 10.1080/01605682.2025.2485263.

[33] Nazerke Baisholan, J Eric Dietz, Sergiy Gnatyuk, Mussa Turdalyuly, Eric T Matson, and Karlygash Baisholanova. "FraudX AI: An Interpretable Machine Learning Framework for Credit Card Fraud Detection on Imbalanced Datasets". In: *Computers* 14.4 (2025), p. 120. DOI: 10.3390/computers14040120.

[34] Andres Alonso-Robisco and José Manuel Carbó. "Should We Trust the Credit Decisions Provided by Machine Learning Models?" In: *Computational Economics* (2025), pp. 1–30. DOI: 10.1007/s10614-025-10855-x.

[35] Thanh Thuy Do, Golnoosh Babaei, and Paolo Pagnottoni. "Explainable machine learning for credit risk management when features are dependent". In: *Measurement: Interdisciplinary Research and Perspectives* 22.4 (2024), pp. 315–340. DOI: 10.1080/15366367.2023.2261186.

[36] Lu Bai and Xuezhou Wen. "Improved Artificial Bee Colony Algorithm for Feature Selection to Enhance the Prediction of Credit Risk in SMEs". In: *Computational Economics* (2025), pp. 1–34. DOI: 10.1007/s10614-025-10955-8.

[37] Murat Doğan, Muhammed Aslam Chelery Komath, and Özlem Sayilir. "Credit rating prediction with ESG data using data mining methods". In: *Future Business Journal* 11.1 (2025), p. 79. DOI: 10.1186/s43093-025-00490-1.

[38] Idowu Aruleba and Yanxia Sun. "An improved Ensemble Method with Data Resampling for Credit Risk Prediction". In: *IEEE Access* (2025). DOI: 10.1109/ACCESS.2025.3563432.

[39] Gang Yao, Xiaojian Hu, Pingfan Song, and Yue Zhang. "A novel class-imbalance-oriented feature selection method based on BPNNs and AdaBoost for enterprise credit risk prediction in the supply chain context". In: *International Journal of Information Technology & Decision Making* (2025). DOI: 10.1142/S0219622025500403.

[40] Ruilin Liang, Yuke Hou, Xin Liang, Bo Li, and XiaoXuan Yao. "The Application of Data Mining Models in Personal Credit Risk Control". In: *International Journal of Image and Graphics* (2025), p. 2750034. DOI: 10.1142/S0219467827500343.

[41] Sichong Lu, Xiaoming Zhang, Yi Su, Xiaojun Liu, and Lean Yu. "Efficient multimodal learning for corporate credit risk prediction with an extended deep belief network". In: *Annals of Operations Research* (2025), pp. 1–38. DOI: 10.1007/s10479-025-06612-w.

[42] Li Zeng and Wee-Yeap Lau. "Nexus Between ESG Performance and Credit Risk in Chinese FinTech Companies". In: *Asia-Pacific Financial Markets* (2025), pp. 1–29. DOI: 10.1007/s10690-025-09528-4.

[43] Guanglan Zhou and Shiru Wang. "Enhancing Credit Risk Decision-Making in Supply Chain Finance with Interpretable Machine Learning Model". In: *IEEE Access* 13 (2025), pp. 14239–14251. DOI: 10.1109/ACCESS.2025.3530433.

[44] Nazário Augusto de Oliveira and Leonardo Fernando Cruz Basso. "Advancing Credit Rating Prediction: The Role of Machine Learning in Corporate Credit Rating Assessment". In: *Risks* (2025). DOI: 10.3390/risks13060116.

[45] Wangke Lin and Yue Liu. "Deep Learning-Based Attention Mechanism Algorithm for Blockchain Credit Default Prediction". In: *International Journal of Advanced Computer Science & Applications* 16.2 (2025).

[46] P Sundaravadivel, R Augustian Isaac, D Elangovan, D KrishnaRaj, VV Rahul, and R Raja. "Optimizing credit card fraud detection with random forests and SMOTE". In: *Scientific Reports* 15.1 (2025), pp. 1–12. DOI: 10.1038/s41598-025-00873-y.

[47] Kianeh Kandi and Antonio Garcia Dopico. "Evaluating the Performance of Deep Convolutional Neural Networks and Support Vector Regression for Creditworthiness Prediction in the Financial Sector". In: *Inteligencia Artificial* 28.76 (2025), pp. 66–84. DOI: 10.4114/intartif.vol28iss76pp66-84.

[48] Lu Wang, Zecheng Yu, Jingling Ma, Xiaofang Chen, and Chong Wu. "A Two-Stage Interpretable Model to Explain Classifier in Credit Risk Prediction". In: *Journal of Forecasting* (2025). DOI: 10.1002/for.3288.

[49] Gustavo EAPA Batista, Ronaldo C Prati, and Maria Carolina Monard. "A study of the behavior of several methods for balancing machine learning training data". In: *ACM SIGKDD explorations newsletter* 6.1 (2004), pp. 20–29. DOI: 10.1145/1007730.100773.

[50] Nitesh V Chawla, Kevin W Bowyer, Lawrence O Hall, and W Philip Kegelmeyer. "SMOTE: synthetic minority over-sampling technique". In:

Journal of artificial intelligence research 16 (2002), pp. 321–357. DOI: 10.1613/jair.953.

[51] Yao Zou and Guanghua Cheng. "Level-Wise Feature-Guided Cascading Ensembles for Credit Scoring". In: *Symmetry* 17.6 (2025), p. 914. DOI: 10.3390/sym17060914.

[52] Gustavo EAPA Batista, Ana LC Bazzan, Maria Carolina Monard, et al. "Balancing training data for automated annotation of keywords: a case study." In: *Wob* 3 (2003), pp. 10–18.

[53] Idowu Aruleba and Yanxia Sun. "Enhanced credit risk prediction using deep learning and SMOTE-ENN resampling". In: *Machine Learning with Applications* 21 (2025), p. 100692. DOI: 10.1016/j.mlwa.2025.100692.

[54] Nurafni Damanik and Chuan-Ming Liu. "Advanced Fraud Detection: Leveraging K-SMOTEENN and Stacking Ensemble to Tackle Data Imbalance and Extract Insights". In: *IEEE Access* (2025). DOI: 10.1109/ACCESS.2024.

[55] Erwis Melchor-Pérez, José Fco. Martínez-Trinidad, J. Ariel Carrasco-Ochoa, and Agustin Santiago-Alvarado. "Oversampling and instance selection to predict credit risk". In: *International Journal of Applied Pattern Recognition* (2025). in press. DOI: 10.1504/IJAPR.2025.10073009.

[56] Mimi Mukherjee and Matloob Khushi. "SMOTE-ENC: A novel SMOTE-based method to generate synthetic data for nominal and continuous features". In: *Applied system innovation* 4.1 (2021), p. 18. DOI: 10.3390/asi4010018.

[57] Leonardo Canete-Sifuentes, Raul Monroy, and Miguel Angel Medina-Perez. "FT4cip: A new functional tree for classification in class imbalance problems". In: *Knowledge-Based Systems* 252 (2022), p. 109294. DOI: 10.1016/j.knosys.2022.109294.

[58] Z Huang. "Clustering Large Data Sets With Mixed Numeric And Categorical Values," Proceedings Of 1st Pacific-Asia Conference on Knowledge Discouvery And Data Mining". In: (1997).

[59] Zhexue Huang. "Extensions to the k-means algorithm for clustering large data sets with categorical values". In: *Data mining and knowledge discovery* 2.3 (1998), pp. 283–304. DOI: 10.1023/A:1009769707641.

[60] Duy-Tai Dinh, Tsutomu Fujinami, and Van-Nam Huynh. "Estimating the optimal number of clusters in categorical data clustering by silhouette coefficient". In: *International Symposium on Knowledge and Systems Sciences*. Springer. 2019, pp. 1–17. DOI: 10.1007/978-981-15-1209-4_1.

[61] Dendy Arizki Kuswardana, Dwi Arman Prasetya, Trimono Trimono, and I Gede Susrama Mas Diyasa. "Comparison of Elbow and Silhouette Methods in Optimizing K-Prototype Clustering for Customer Transactions". In: *Jurnal Ilmiah Edutic: Pendidikan dan Informatika* 12.1 (2025), pp. 43–48. DOI: 10.21107/edutic.v12i1.29744.

[62] Octavio Loyola-González, Miguel Angel Medina-Pérez, José Fco Martínez-Trinidad, Jesús Ariel Carrasco-Ochoa, Raúl Monroy, and Milton García-Borroto. "PBC4cip: A new contrast pattern-based classifier for class im-

balance problems". In: *Knowledge-Based Systems* 115 (2017), pp. 100–109. DOI: 10.1016/j.knosys.2016.10.018.

[63] Hans Hofmann. *Statlog (German Credit Data)*. UCI Machine Learning Repository. DOI: https://doi.org/10.24432/C5NC77. 1994.